Mechanics – N T Hong Van

1. A particle moves through a circular are of length l and radius r in time t. Which expression gives its angular velocity about the centre of the circle?

   A. $\dfrac{l}{rt}$

   B. $\dfrac{r}{lt}$

   C. $\dfrac{2\pi}{lt}$

   D. $\dfrac{2\pi r}{t}$

2. The axis of rotation of a circular plate is through its centre and perpendicular to its plane. Two particles P and Q are on the plate. The distance of P from the axis is twice that of Q. At a given instant what is the value of

$$\frac{\text{the linear velocity of P}}{\text{the linear velocity of Q}} \ ?$$

   A. 4

   B. 2

   C. 1/2

   D. 1/4

3. A communication satellite is moving at constant speed in a circular orbit about the Earth. Which row is the table is correct about the momentum and kinetic energy of the satellite?

	momentum	*kinetic energy*
A	constant	changing
B	constant	constant
C	changing	changing
D	changing	constant

Solution: D

4. Which statement is correct about a particle moving in a horizontal circular path with constant angular velocity?

   A. The linear momentum is constant but the kinetic energy varies

   B. The kinetic energy is constant but the linear momentum varies

   C. Both kinetic energy and linear momentum are constant

   D. Both speed and linear velocity are constant

5. A body is moving in uniform circular motion. Which row in the table correctly describes the linear velocity, angular velocity and linear acceleration of the body?

	linear velocity	*angular velocity*	*linear acceleration*
A	constant	constant	varying
B	constant	constant	zero
C	constant	varying	constant
D	varying	constant	varying

Solution: D

6. What is the mean angular speed of the minute hand of a clock?

   A. 1.4×10^{-4} rad/s

   B. 1.7×10^{-3} rad/s

   C. 5.2×10^{-3} rad/s

   D. 1.0×10^{-1} rad/s

7. A spacecraft is moving in a circular orbit around the Earth when its engine is fired and produces a force on the spacecraft exactly equal and opposite to that exerted by the Earth's gravitational field. How would the spacecraft move?

   A. along a spiral path towards the Earth's surface

   B. along the line joining it to the centre of the Earth (i.e.radially)

   C. along a tangent to the orbit

   D. in a circular orbit with a longer period

8. An engine has a component rotating at 3000 revolutions per minute in a circle of radius 8.0 cm. Which of the following is its centripetal acceleration?

   A. 25 m s^{-2}

   B. 7900 m s^{-2}

   C. 3.1 x 10^4 m s^{-2}

   D. 7.2 x 10^7 m s^{-2}

9. An oscillating pendulum bob of mass 1.27 kg is suspended from a string of length 0.600 m. At the instant when the string is vertical, it is moving with a velocity of 0.575 m/s horizontally. Which of the following is the tension in the string at this instant?

   A. 11,8 N

   B. 12,5 N

   C. 13,2 N

   D. 13,7 N

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10. A swinging pendulum bob of mass m and weight W is suspended from a string of length r. At the instant when the string is vertical, tension in the string is T and the bob is moving at speed v. Which of the following expressions relating W and T is correct?

    A. $T = W$

    B. $T - W = v^2/r$

    C. $T - W = mv^2/r$

    D. $W - T = mv^2/r$

11. A car is travelling at a constant speed v over a humpback bridge with radius of curvature r. The car of mass m remains in contact with the road and exerts a net force R on the road when its is at the top of the bridge. Which expression gives the correct value for R?

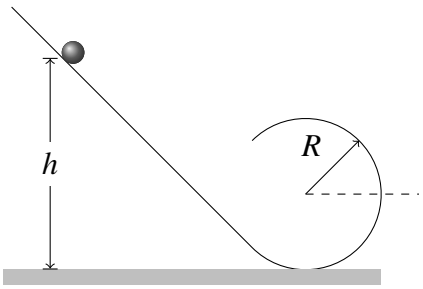
    A. R = mg + mv^2/r

    B. R = mv^2/r

    C. R = mg - mv^2/r

    D. R = mv^2/r - mg

12. A body is released from height h along an inclined plane. For what minimum height, the body will complete revolution in the vertical circle of radius R



    A. 6.0R

    B. 4.0R

    C. 2.5R

    D. 2.0R

Solution: For completing motion in a vertical circle, the velocity at the lowest point is equal to $\sqrt{5Rg}$. Potential energy = Kinetic energy

$$mgh = \frac{1}{2}m\left(\sqrt{5gR}\right)^2$$

$$\rightarrow h = \frac{5}{2}R = 2.5R$$

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